CHRISTOPHER A. RIVAS

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EDUCATION

Drexel University College of Medicine, Philadelphia, PA — *expected* May 2029
Doctor of Medicine

Texas Christian University, Fort Worth, TX — May 2025
Master of Science, Neuromuscular Physiology

Texas Christian University, Fort Worth, TX — May 2023
Bachelor of Science, Kinesiology, Movement Science
Awards: Cum Laude (2023); Dean's Honor List and Scholar (2021)

PUBLICATIONS

Rivas, C. A., Voskuil, C. C., Hahs-Vaughn, D. L., Stock, M. S., & Carr, J. C. (2025). Motor Unit Activity During Maximal-Intent Dynamic Muscle Actions Varies by Intensity and Sex In Healthy Adults. *Journal of Neurophysiology*, jn.00184.2025. https://doi.org/10.1152/jn.00184.2025

Fasih, M., Voskuil, C., Rivas, C., Lee, B., Nadolny, C., Dietrich, L., & Carr, J. (2025). Rectus femoris and vastus intermedius muscle asymmetries following ACL reconstruction. International Journal of Sports Medicine, a-2604-7425. https://doi.org/10.1055/a-2604-7425

PRESENTATIONS

Drexel University Discovery Day — 2026

Global Neuroscience Institute Chiari Support Group — 2026

TCU Master's Thesis Defense — 2025

TCU Harris College Student Research Symposium — 2023

TCU Student Research Symposium — 2021

RESEARCH EXPERIENCE

Research Assistant, DUCOM Neurosurgery Department, Philadelphia, PA — 2025-Present

- Assisting with IRB drafting, study design, data collection/analysis, and manuscript development for projects on advancing Chiari malformation management and treatment. Supervised by Dr. Erol Veznedaroglu.
- Assisting with IRB drafting, study design, data collection/analysis, and manuscript preparation for development of a prognostic grading instrument for mechanical thrombectomy candidacy in acute ischemic stroke due to large vessel occlusion. Supervised by Dr. Mandy Binning.
- Assisted in the large, multi-institutional data collection/analysis of the microsurgical and endovascular operative management and outcomes of ruptured aneurysms. Supervised by Dr. Tyler Scullen.

Graduate Research Assistant, Neuromuscular Physiology Research Lab, TCU — 2023-2025

- Applied surface electromyography to analyze motor unit control in the novel setting of dynamic muscle actions, providing new evidence of load-dependent motor unit firing behavior and divergent firing characteristics between sexes during maximal-intent dynamic muscle actions. Supervised by Dr. Joshua Carr

Research Assistant, Neuromuscular Physiology Research Lab, TCU 2021-2023

- Assisted with the development and data collection/analysis of ultrasound imaging and electromyography across research studies in areas such as Parkinson's Disease, Cross-Education, and neuromuscular control. Supervised by Dr. Joshua Carr.

Research Assistant, Exercise Physiology Research Lab, TCU 2020-2021

- Assisted with development and data collection about the relationship between nutrient intake and athletic performance and recovery in elite athletes. Supervised by Dr. Mena Shah.

CLINICAL EXPERIENCE

Scribe and Intern, Longhorn Brain and Spine, Fort Worth, TX 2023-2025

- Under Dr. Grant Booher, assisted in preparing charts, rooming patients, and documenting the patient's chart. Developed an understanding of the imaging interpretation process (X-rays, MRI, CT scans, etc.) and the medical and surgical management of neurosurgical patients.

Intern, Southwest Sport and Spine, Fort Worth, TX 2023-2024

- Under Dr. Robert Menzies, MD, observed and assisted with patient intake, documentation, and procedure support in acute and chronic pain management.

LEADERSHIP AND MENTORSHIP EXPERIENCE

Research Chair, DUCOM Chapter of American Association of Neurological Surgeons 2026-Present

- Responsible for connecting members of the chapter with neurosurgical research opportunities and serve as a point of contact for all research needs and advice.

Ambassador, Harris College of Nursing and Health Sciences, Texas Christian University 2023-2025

- Selected to represent Harris College at various events by assisting in preparing for events, welcoming prospective students, offering them guidance through the application process, and supporting their college experience.

Teaching Assistant, Texas Christian University 2022-2025

- Assisted the professor in instructing various lectures and/or laboratory experiments while attending to the development and needs of the students.
- Taught across: neurophysiology, exercise physiology, general and organic chemistry, and anatomy and physiology

Tutor 2021-Present

- Tutored students 1:1 in various undergraduate and graduate courses by creating tailored study plans, problem sets, and study guides.

VOLUNTEERING

Volunteer, Tarrant County Food Bank**,** Texas Christian University, Fort Worth, TX 2023-2025

- Distributed food to families in need across the Tarrant County area. Over 15,000 families served, with an impact value of over $3,000.

MEMBERSHIPS

Member, American Association of Neurological Surgeons 2026-Present
Member, Congress of Neurological Surgeons 2025-Present